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Applications of Solid, Slow-Release Scale Inhibitors in Fractured Reservoirs

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Abstract

Regardless of whether a well is being developed onshore or offshore, inorganic scale is a persistent challenge in oilfield operations, and this challenge is worsened by the use of seawater or reuse of high salinity produced brines. Thus, scale control programs are a standard part of most wells. Liquid scale inhibitors are most common, but alternatives are available in the form of solid, slow-release inhibitors. While liquid products lose their efficacy as soon as they are produced from the well, slow-release solids can provide much longer periods of protection. If properly selected and applied, it is possible to see multiple months of protection against scale formation. This paper will discuss multiple case studies describing how these products can be successfully used. Focus will be given to a family of products that combine high activity, extended release, and favorable cost-to-treat.

In this paper, we will discuss initial laboratory results used to screen and compare the performance of various commercially available products. In some cases, the release behavior of a given composition is achieved by controlling the chemical composition of an active molecule; in other cases, the release behavior is achieved by controlling the physical properties of a composition (for example, through the use of encapsulation or barrier materials). Differences in release behavior are apparent, and although direct field comparisons are difficult to make, we anticipate these differences will translate to differences in performance in downhole environments. Field results will be discussed, demonstrating that the utility of these products extends beyond laboratory testing.

In a representative 6-well trial in the Bakken reservoir, North Dakota, scale protection was achieved for approximately 10 months as indicated by monitoring returns in the produced fluids. A separate 6-well case study in the Uinta Basin, Utah is forecast to yield similar results. Downhole equipment prone to scale buildup and eventual failure was kept online and scale-free for multiple months, indicating effective protection by the solid slow-release treatment.

Slow-release scale inhibitors have been available on the market for many years, but it can be challenging to achieve consistently positive results that provide confidence that a well is protected. A correct understanding of the scope and limitations of these products is an important first step, and this paper will help provide that foundation. Our goal is to demonstrate that effective slow-release scale inhibitor treatments are possible, but the treatment strategy needs to properly integrate downhole conditions, a product's properties and behavior and a target protection profile (e.g., length of protection, type of scale, etc.).

Introduction

Inorganic scale (see [Figure 1](#)) has been recognized for decades as a serious challenge to efficient oilfield operations ([Vetter 1976](#)) and its economic impacts can be severe ([Graham 2005](#)). Accordingly, there have been numerous descriptions in the literature of various chemicals that can be used to preemptively mitigate the formation of scale in oilfield production environments ([Olajire 2015](#)). The preferred approach for addressing scale formation is to use threshold inhibitors, or kinetic inhibitors, which function at substoichiometric concentrations, relative to the concentration of scaling species present in a wellbore environment. Mechanistically, these molecules function by inhibiting nucleation of scaling species, retarding crystal growth after nucleation has occurred, inhibiting deposition onto wellbore surfaces or topside equipment, or combinations thereof ([Sorbie 2004](#)). The effectiveness of a particular inhibitor can vary, however, depending on the type of scale that forms, how aggressively that scale forms, the overall composition of the brine, and the test conditions used to evaluate the inhibitor (e.g., time, temperature) ([Fraser 2016](#)). Thus, it is not always immediately apparent what might be a preferred chemistry for a particular application, and some type of product screening is performed prior to field application. Once a preferred inhibitor chemistry has been identified, the treatment can then be applied to a well using continuous injection, batchwise injection, or via a squeeze treatment ([Kan 2020](#)). Each of these approaches has its own advantages and disadvantages, but when used correctly, all can be effective.



Figure 1—Inorganic scale formation in an unstable brine with varying dosages of a chemical scale inhibitor (increasing left to right). Effective inhibition is observed at higher concentrations as indicated by the lack of precipitate formation.

An alternative approach to addressing oilfield scale, particularly in the early stages of a well's life, is to use a solid slow-release scale inhibitor. Typically, solid scale inhibitors are included with the proppant in hydraulic fracture stimulations. The slow-release behavior is particularly important in unconventional wells where the majority of the water the well will ever make occurs within the first year ([Haines 2024](#)). While liquid scale inhibitors are fully soluble products that only provide short periods of protection (with the exception of squeeze treatments), solid products are generally designed to release an active scale inhibiting species over a prolonged period. Many different approaches have been explored over the years using various approaches such as encapsulation of a solid active ingredient with an inert, low permeability coating (encapsulation) ([Saldungaray 2015](#)), infusion of an active ingredient into an inert matrix ([Gupta 2013](#)), chemical modification of the active ingredient ([Dreyer 2021](#)), or combinations thereof (see [Figure 2](#)).

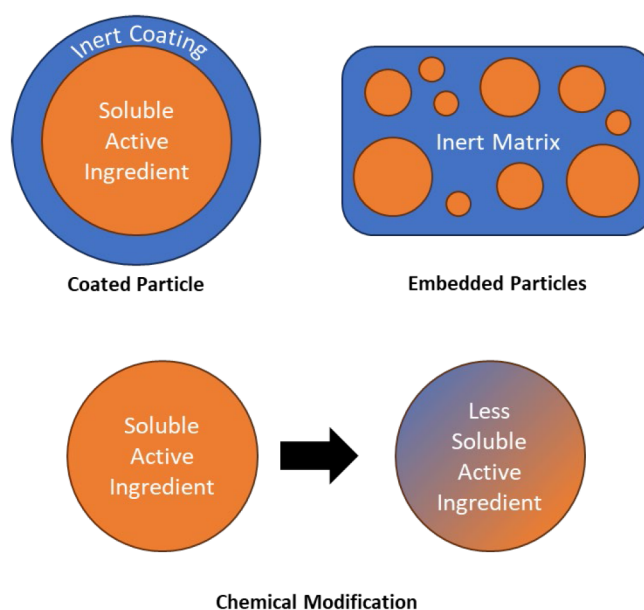


Figure 2—Schematic overview of approaches for converting a soluble, liquid-phase scale inhibitor into a solid, slow-release form.

Historically these products were designed with specific production environments in mind and thus the active chemistries, overall activity levels, release profiles and other physical properties (e.g., particle size, mechanical strength) vary widely. There can be consequences to some of the formulation choices that were made, however. For example, while the incorporation of inert matrix materials (e.g., diatomaceous earth) or polymeric coatings (e.g., PVdC) slow the release of the active ingredient, they also lead to the possibility of formation damage because these materials do not degrade under downhole conditions. Similarly, the use of porous proppants as an inert matrix can be beneficial because the material retains the mechanical strength required of proppant particles, the overall activity level of the finished product can be very low (typically range from 5-25%) making this a costly treatment method.

Despite these differences, all share the common end goal of medium to long term mitigation of scale formation in downhole environments and associated surface equipment. Using a combination of laboratory testing and field trial evaluations, we describe here a family of products that fills a similar role, while seeking to address some of the problems associated with other solid scale inhibitors that have been previously described.

Experimental Methods

Static Bottle Testing

Evaluation of the scale inhibition properties of the products was performed by testing fully soluble liquid analogues of the active ingredients used to formulate the solid slow-release forms of the product. Preliminary screening studies were performed using static bottles similar to methods that have been described previously (NACE TM0374-2016). Tests were performed at 50 mL scale at a temperature of 194 °F (90 °C) for a period of 24 h in calcite-, barite-, or gypsum-forming brines as indicated. The presence or absence of scale was monitored visually as shown in the results.

Dynamic Scale Loop Testing

The liquid analogues were also tested using a dynamic scale loop (DSL) apparatus similar to the configurations that have been described previously in the literature (Bazin 2005, NACE TG 072 2005). A synthetic brine with a moderate overall salinity (TDS = 105,320 ppm) and moderately aggressive calcite

scaling index (12.53) was used for all the DSL tests. Tests were performed at a temperature of 194 °F (90 °C) and a combined flow rate of 2.0 mL/min. A failing result was defined as an increase in pressure of approximately 6-7 PSI relative to the initial baseline pressure, while a passing result was defined as a lack of a substantial pressure increase for at least 2x the time required for the blank (uninhibited) brine to fail.

Release Behavior Testing

The release behavior of the compositions described herein was evaluated using static bottle tests. Approximately 1 g of the solid scale inhibitor was placed in a bottle containing 100 mL of demineralized water and the bottle was sealed and placed in an oven preheated to a temperature of 90 °C. At the indicated times, the solids in the bottle were removed via filtration and the phosphonate concentration in the filtration was quantified using a Hach spectrophotometer (Method 8007). The solids were dried, weighed and placed back into the bottle with a fresh 100 mL of demineralized water. Release rate was quantified in terms of the phosphonate concentration as a function of time.

Results

Laboratory Testing

As described previously ([Dreyer 2025](#)), we have commercialized two forms of a solid slow-release scale inhibiting composition, and both have been explored extensively in the laboratory and in the field. Both versions of the product comprise an organophosphonate active ingredient that has been rendered sparingly soluble via chemical modification of the active molecule (see [Figure 2](#), bottom). A key advantage of using this formulation is that the overall activity of the product remains very high (>95%), in contrast to other slow-release scale inhibitors that have much lower levels of activity. A separate version of the product also contains an inert polymer coating on the surface. The activity of this product is moderately reduced (~85%), but the addition of the coating provides an additional barrier to release of the active molecule into the downhole fluid environment, particularly at the early stages of the well's life. Once the coating is compromised and water permeates into the core containing the active molecule, the coating's usefulness is significantly reduced and the subsequent extended release stems from the sparing solubility of the active molecule itself. We refer to these two versions as either the live (unencapsulated) or encapsulated form of the product.

Building on this foundation, we have continued to optimize the product, particularly regarding its ability to inhibit scale at low concentration and its applicability toward many different types of scale. The previous live and encapsulated version of the product contained a mixture of active ingredients (referred to herein as Active 1). As described in our previous work ([Dreyer 2025](#)), the blend of these active ingredients proved more effective at inhibiting scale than either of the active ingredients alone. In the present work, we have shifted toward using another organophosphonate active ingredient (referred to herein as Active 2) that provides better scale protection than Active 1, as indicated by a reduced minimum inhibitor concentration (MIC) as will be shown in the results that follow. In addition to providing equivalent or better scale protection than our previous versions of the product, reformulation with the new active ingredient has simplified our manufacturing process and reduced overall costs.

The new form of the product is prepared exclusively as a live (unencapsulated) form with a nominal size range of 30-100 mesh (see [Figure 3](#)). We have selected this size range to maximize compatibility with proppants commonly used during completion operations, but the size can be adjusted to accommodate other proppant sizes or application needs. With regards to focusing on the live version of the product, there can be circumstances where encapsulation is valuable, but we have generally found that it is unnecessary for most applications for three primary reasons:

1. Due to advancements in altering the chemical composition of the unencapsulated form, the dissolution profile can be tuned to match the encapsulated form, rendering the encapsulation material unnecessary.
2. The application of a coating adds significant cost, primarily because specialized equipment and techniques are needed to apply uniform coatings to small particles of this type.
3. Coating levels vary widely (generally in the range of 10-40 wt%), but regardless of the amount of coating applied, it is an inert material that will dilute the concentration of the active ingredient being coated.

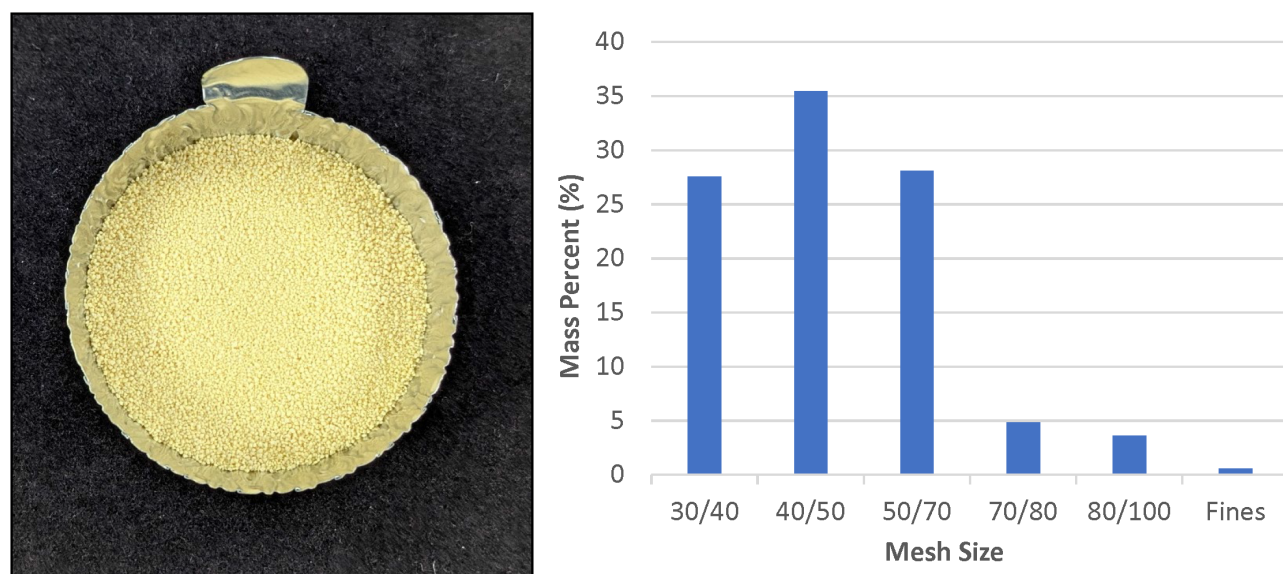


Figure 3—The solid slow-release scale inhibiting composition described herein (left) and its particle size distribution (right).

From a practical perspective, it is very challenging to directly measure the scale inhibition properties of solid slow-release products like those described above. The sparing (but non-zero) solubility leads to continually changing and difficult-to-control concentrations of the active ingredient. While this is desirable for a field application where the continuous introduction of water (both flowback from the completion operation and connate water from the formation) into the wellbore leads to ongoing dissolution of the active molecule, it is problematic for lab tests that are typically used to evaluate scale inhibitor performance. Thus, for the purposes of this study, fully soluble liquid analogues comprising the same active ingredients were used to circumvent this challenge.

The slow-release product we have formulated contains a highly effective organophosphonate active ingredient. The motivation for this choice is supported by the data shown in Figures 4–6 below. The relative efficacy of Active 1 and Active 2 were compared in three separate brines using static bottle tests performed at moderately elevated temperature (194 °F). Both liquid analogues were tested using active-normalized concentrations, so the results shown in the figures allow for direct comparison of the two scale inhibitors. Simple visual detection was used to determine the presence or absence of scale.

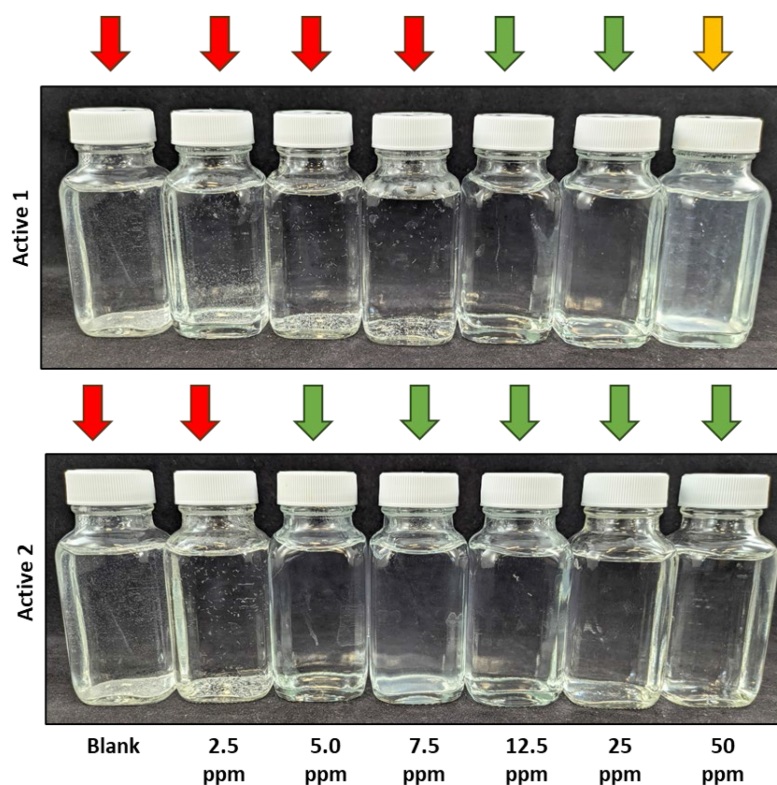


Figure 4—Static bottle test results after 24 h for the previously used active ingredient (Active 1; top row) and the improved active ingredient (Active 2; bottom row) in a calcite-forming brine.

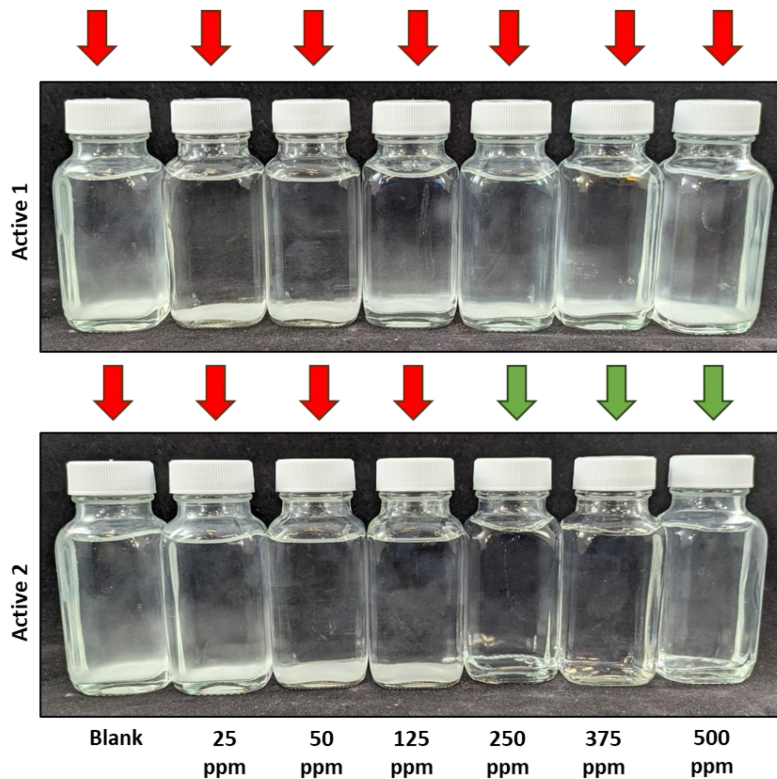


Figure 5—Static bottle test results after 24 h for the previously used active ingredient (Active 1; top row) and the improved active ingredient (Active 2; bottom row) in a barite-forming brine.

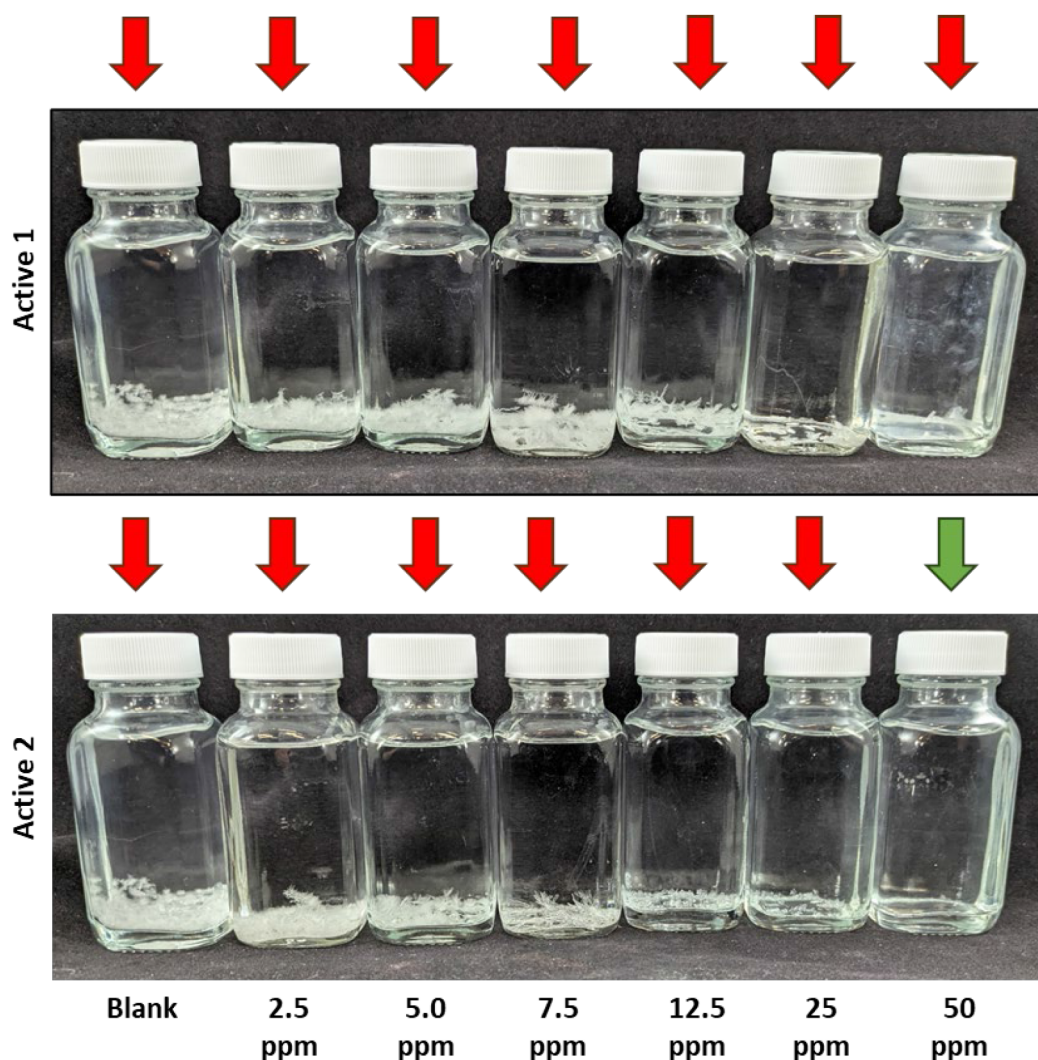


Figure 6—Static bottle test results after 24 h for the previously used active ingredient (Active 1; top row) and the improved active ingredient (Active 2; bottom row) in a gypsum-forming brine.

In the calcite-forming brine (Figure 4), Active 1 exhibited an MIC of approximately 12.5 ppm, while Active 2 exhibited an MIC of approximately 5.0 ppm. Moreover, at high concentration (50 ppm), Active 1 showed signs of incompatibility or overdosage in this particular brine. The barite-forming brine used in this test was considerably more aggressive than either the calcite- or gypsum-forming brines, so much higher product dosages were needed to see any effect with Active 1 or Active 2. In this brine (Figure 5), Active 1 did not display acceptable inhibition at any concentration in the dosage range tested (25–500 ppm), indicating poor efficacy in such an aggressive brine. Conversely, Active 2 exhibited an MIC of approximately 250 ppm. In the gypsum-forming brine (Figure 6), Active 1 again failed to display acceptable inhibition at any concentration in the dosage range tested (2.5–50 ppm). Active 2 exhibited an MIC of approximately 50 ppm. Collectively, these results indicate that Active 2 is a more effective scale inhibitor and we anticipate that reformulating our slow-release product to incorporate Active 2 will lead to improved performance by reducing the required MIC.

The static bottle tests were complemented by a series of DSL tests to confirm that the favorable results from the extended length (24 h) bottle tests would be mirrored in much shorter scale loop measurements. In this series of tests, a moderately aggressive calcite scaling brine was pumped through a narrow ID stainless steel tube at a combined flow rate of 2.0 mL/min. The blank (uninhibited) brine (orange line) showed measurable scale deposition after approximately 17 minutes, as indicated by a spike in the differential

pressure measured across the tube (see Figure 7). Actives 1 (blue lines) and 2 (green lines) were tested at concentrations ranging from 2.5-7.5 ppm. Both active ingredients failed at a concentration of 2.5 ppm (dotted lines). At 5 ppm (dashed lines), Active 1 still failed (albeit moderately slower than what was observed at 2.5 ppm) while Active 2 showed no significant increase in pressure. Both Active 1 and Active 2 showed a passing, stable pressure response at 7.5 ppm (solid lines). The MIC required to prevent scale deposition will vary depending on the water composition, test conditions (pressure and temperature), and active scale inhibitor, but for this particular combination of active ingredients tested under these conditions, we can say the MIC of Active 1 is approximately 7.5 ppm while the MIC of Active 2 is approximately 5 ppm. This provides further support for the reformulation of our slow-release solid product to include Active 2 instead of Active 1.

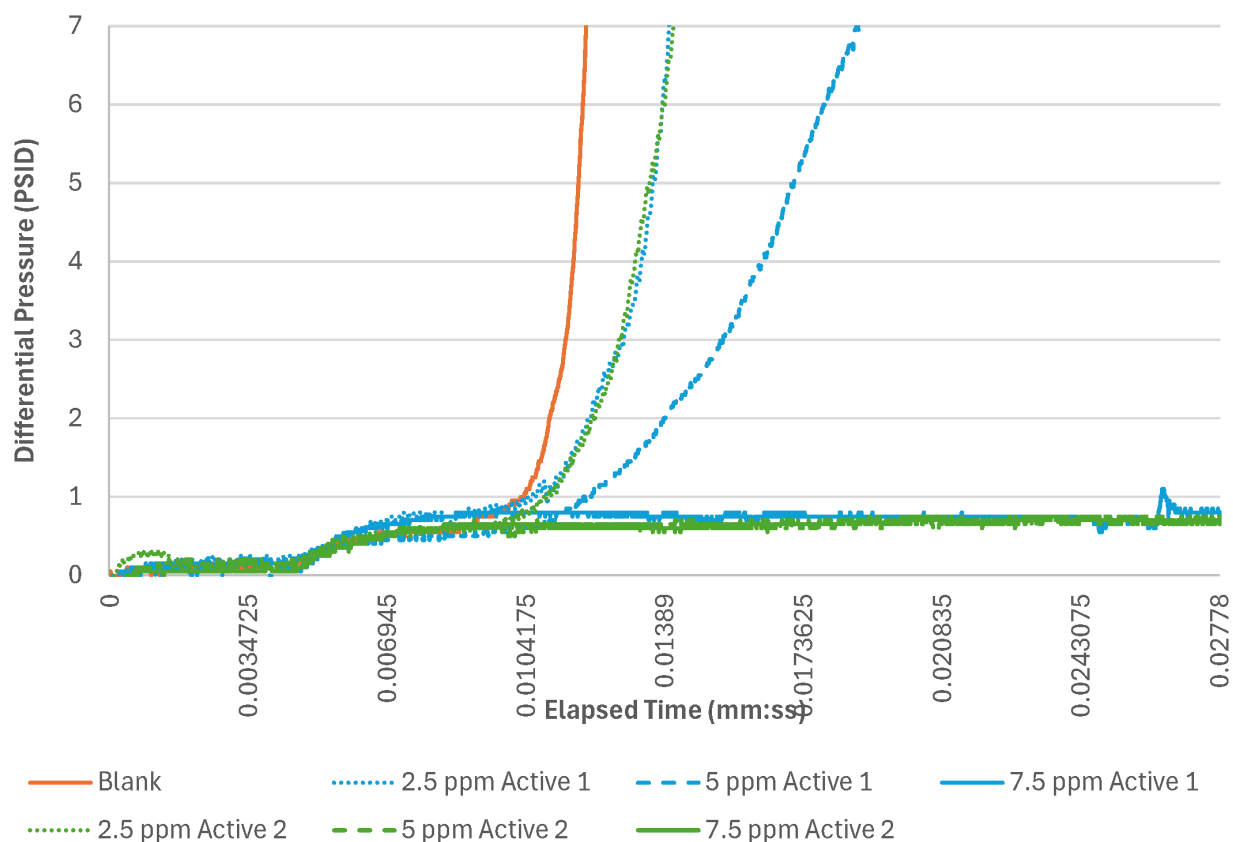


Figure 7—Dynamic scale loop test results for the active ingredients contained in the slow-release composition described herein (Active 2) referenced against the ingredients in our previous version of the product (Active 1) in a model Permian brine.

It is essential that a slow-release solid scale inhibitor demonstrate an ability to inhibit scale at low dosages, as we have shown, but it is also essential that any product used for such an application do so for a prolonged period. The desired release profile for a slow-release product can vary considerably depending on individual well conditions, and therefore it is not possible to define standardized test conditions. For the purposes of this testing, we have evaluated the release rate of the reformulated composition at a low concentration (1 g per 100 mL) in demineralized water at a moderate temperature (90 °C). Release rate increases with temperature, and will generally drop with increasing salinity, but this can vary depending on how the active ingredient interacts with various ions present in field brines. In our lab tests, the solid was separated from the water via filtration and quantification of the active species in the filtrate was monitored for a prolonged period. The water in the bottles was replaced with a fresh aliquot when each data point was collected in order to simulate the introduction of virgin connate brine from the formation. As shown in Figure 8 below, the concentration of the active ingredient was initially high, indicating a relatively rapid release of active species from the

product. The concentration of active species drops over time, but over the course of the approximately 2.5-month test shown here, the measured value remains above what would likely be a required MIC for most scaling environments. The exact time required for the active concentration to fall below typical MIC values will vary with test conditions (temperature, agitation, product concentration, etc.), but these results are an excellent indicator that the product can be expected to provide prolonged periods of protection against scale.

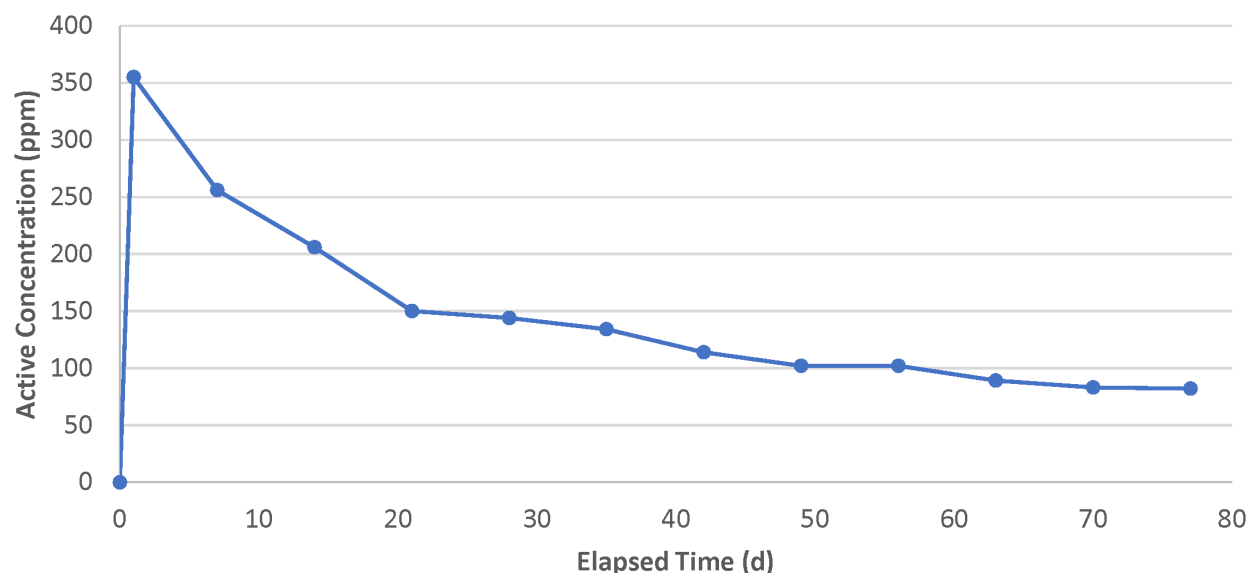


Figure 8—Release rate of the blended active ingredient from the solid slow-release composition described herein.

As noted previously, the predominant mechanism of extending the release profile stems from chemical modification of the organophosphonate active ingredients used to formulate the product. There are many ways to modify these ingredients, but one of the simplest is to vary the composition of the counter-cations that balance the anionic charge of the organophosphonate. Adjusting the type and proportion of cations in the blend can influence the dissolution rate of the final product. In general, certain cation characteristics may lead to slower or faster dissolution, depending on the formulation. This concept is illustrated in the data shown in [Figure 9](#). Two different samples were prepared using Active 2 with varying counter-cations. No coatings, matrices or other inert materials were used in any of these products. Release profiles for the live (unencapsulated) and encapsulated versions of the previous generation of product are shown for comparison. The release profiles of the reformulated products varied significantly, despite the use of a common active ingredient, indicating that it is possible to tune the release behavior of a slow-release solid scale inhibitor to meet the requirements of a specific application.

The products we have commercialized thus far contain a single composition of active ingredients, but we are actively developing new forms that are designed to meet more specific requirements (e.g., low temperature wells vs. high temperature wells, short-term protection vs. long-term protection). This type of formulating can be very application-dependent and is typically handled on a case-by-case basis.

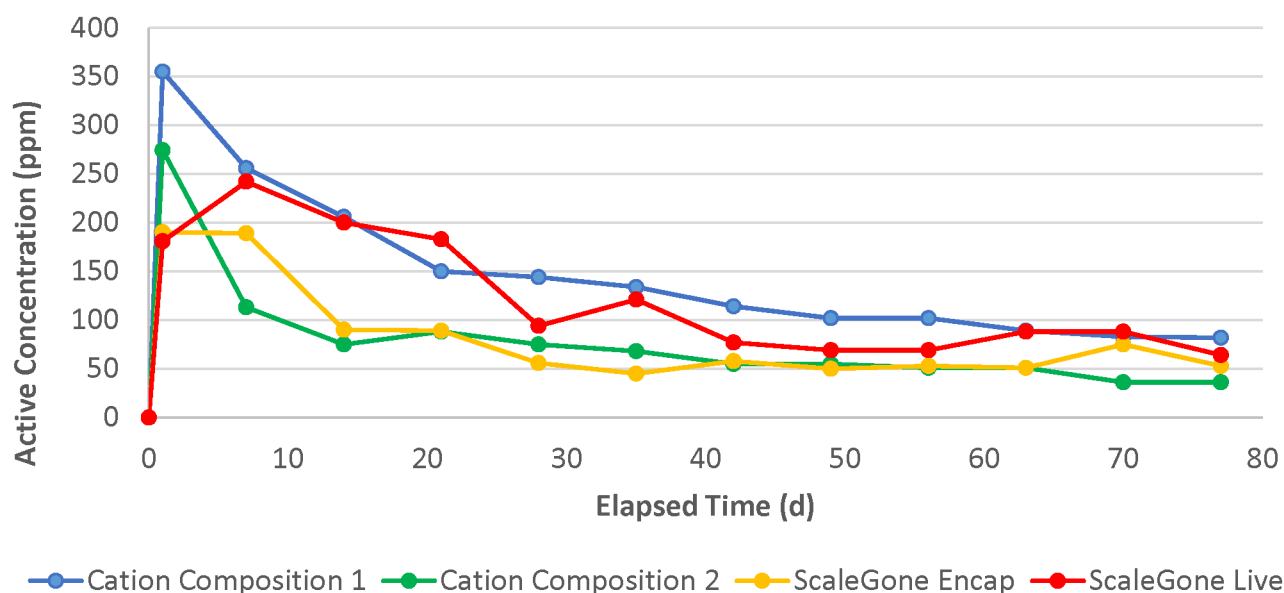


Figure 9—Release behavior of the anionic organophosphonate active ingredients with varying counter-cation composition for the reformulated product (solid lines) and the live and encapsulated versions of the previous generation of product (dashed lines).

The reformulated product comprising Active 2 described above is a new offering and we have not had an opportunity to test it in a field trial. Thus, the field results discussed below are based on the previous versions of the product (both live and encapsulated) that comprised Active 1. The results of the field work using these older products were favorable, as will be described, but we believe the transition to the reformulated product will provide increased protection at a lower cost. The results of those field trials will be discussed in future work.

Bakken Field Trial

A preliminary field trial of the encapsulated form of the product (comprising Active 1) was conducted with a customer in the Bakken formation in 2023. Produced fluids in the Bakken can have very high TDS (often over 250,000 ppm) and are prone to forming a variety of types of scale include calcite (CaCO_3), siderite (FeCO_3), and halite (NaCl) (Cook 2015). For this trial, the product was pumped at a rate of 0.3 pounds per thousand gallons through all stages of a 6-well pad. Laboratory testing conducted before the start of the trial indicated an expected MIC of approximately 0.8 ppm.

Monitoring of the concentration of the active ingredient in the produced fluids began approximately 45 days after the wells' completion and continued for approximately 13 months. As shown in Figure 10, the measured concentration of the active quickly climbed to multi- ppm levels (well above the MIC) in all 6 wells shortly after monitoring started. After 10 months, the active concentration in the flowback of 4 of the 6 wells was still above the expected MIC, but after 12 months, the concentration had fallen below the expected MIC in all 6 wells. This provided a good indication that the slow-release product had effectively been spent and another scale control program (e.g., continuous injection, squeeze treatment, or reapplication of a solid product) would need to be implemented to ensure the wells remained free of scale deposits.

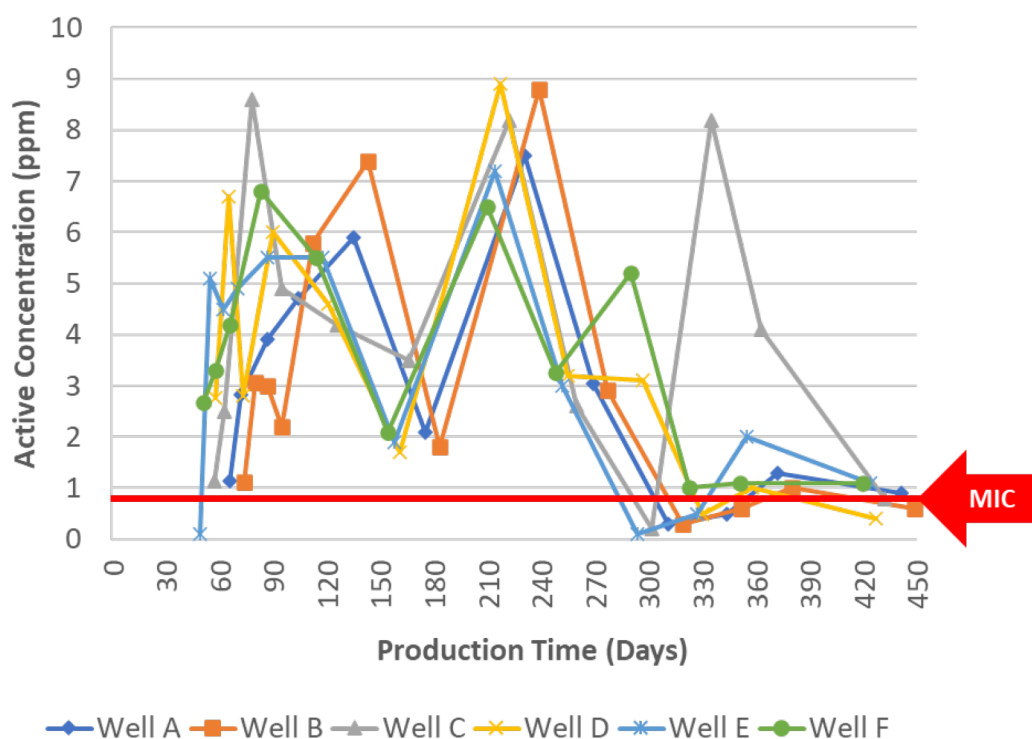


Figure 10—Flowback monitoring results from a 6-well trial conducted in the Bakken.

Uinta Basin Case Study

The Castle Peak formation is a low permeability shale, oil bearing reservoir located in the Uinta Basin, Utah. This formation has recently been developed utilizing modern horizontal multi-stage fracturing techniques (Sigler 2024). The wells have high oil productivity with average 12-month cumulative oil production of 245,000 bbl but also will produce nearly 370,000 bbl of water. Table 1 shows the scaling anions and cations for the produced water. While this produced brine is relatively fresh, it still has an average calcite scaling index of 1.6 which has resulted in scale deposition and periodical failures (Maschio 2007).

Table 1—Castle Peak average produced water analysis.

Species	Concentration (mg/L)
Barium	10
Calcium	200
Magnesium	25
Potassium	90
Strontium	25
Iron	10
Sulfate	50
Bicarbonate	2,000
Dissolved CO ₂	20
<i>TDS</i>	<i>20,000</i>
<i>PH</i> <i>BHT (°F)</i>	<i>8.3</i> <i>200</i>

Utilizing the water analysis above, the live (unencapsulated) version of Active 1 was used for the lab scale inhibition study. A DSL study was run at 200 °F, indicating a minimum inhibitor concentration of approximately 1 ppm. The forecasted water decline rate combined with the release profile of the product was used to design the mass of product to be injected. A total of 4,400 lbs was recommended for each of 6 wells. Internally developed models forecasted that approximately 50% of the product would be consumed after one year of production.

The treatment was designed to place the product during the last 20% of each fracture stage with the goal of leaving the material near the perforations of the wellbore. This was done to increase the likelihood that as water was produced through the hydraulic fractures and into the wellbore, contact with the solid scale inhibitor would be maximized.

The trials are still ongoing, but Figure 11 shows the residual inhibitor concentration to date. Each well is exhibiting a higher initial release rate which then decreases to a steady state concentration. Contrary to the steady lab data release profiles provided, it is not uncommon for field residuals to be more erratic, likely a result of heterogenous flow patterns and varying contributions of production from along the lateral. These wells continue to show good residuals and there is no indication of scaling occurring resulting in expansion of the program.

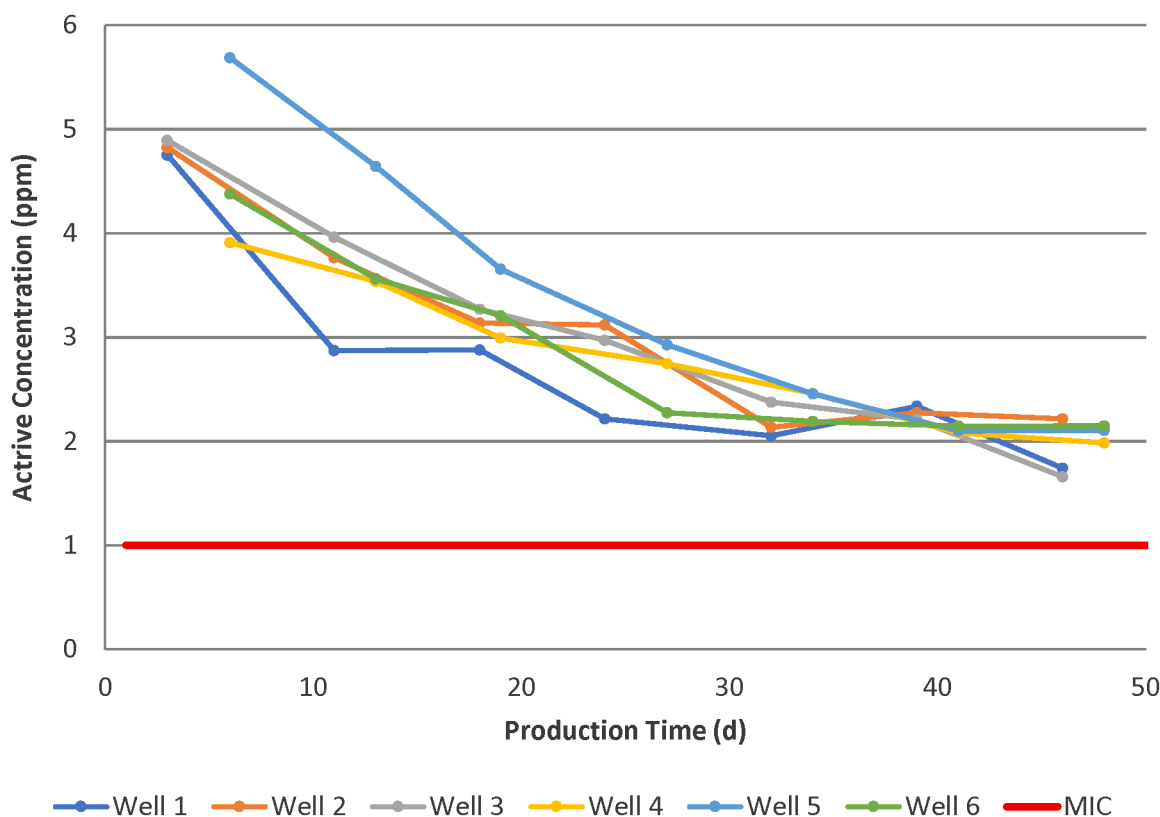


Figure 11—Flowback monitoring results from a 6-well trial conducted in the Castle Peak.

Conclusions

Scale has been and will continue to be a major challenge for oilfield operations, during both well completion and production. A range of effective treatment options exist, but slow-release solid scale inhibitors have been proven to be competent and cost-effective ways to prevent scale formation and deposition. In many cases, it is possible to use moderate dosages while still achieving extended periods of protection (multiple months), highlighting the inherently efficient way these products release active scale inhibitor into the wellbore.

In this paper, we have described a reformulated and improved version of our slow-release product. Laboratory testing on this new product, as well as field applications conducted with its predecessors, indicate the material is highly effective against scale in conditions that are relevant for well operators looking to maximize production from their assets. This patent-pending technology is designed to be applicable for the most common scaling environments encountered in a wide range of unconventional plays, but modifications of the product are possible if more specific challenges need to be addressed.

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